Palaeomagnetic Study on a Granitic Rock Mass with Normal and Reverse Natural Remanent Magnetization

Haruaki ITO

Physics Laboratory, Shimane Agricultural College, Matsue, Shimane-ken
(Read May 13 1963; Received April 1, 1965)

Abstract

The author collected intrusive rocks in Nose district in Kinki Province and measured their natural remanent magnetization. Results of measurements of rocks collected from one rock body are as follows: the natural remanent magnetization of the intrusive rocks collected at the contact zone with country rocks was normal, while that of the intrusive rocks collected at places a little far from the contact zone was reverse. The directions of remanent magnetization were neither typical normal nor reverse. The normal direction deviated easterly relative to the present field direction, and the reverse one westerly.

A fact that the rocks from a single mass possess both the normal and reverse magnetization can be explained, considering that the rock mass had been formed at the times when a transition of the geomagnetic field occurred, and the intermediate direction of magnetization seems to indicate the transition of the dipole of the geomagnetic field. The pole positions obtained from the intermediate direction of natural remanent magnetization lie in the same zone as those obtained from the Mio-Pliocene rocks in southwestern Japan.

1) Introduction

Since 1929 when Matuyama found rocks with reverse natural remanent magnetization (reverse N.R.M.) in Japan and Korea, many investigators have studied on the reverse N.R.M. of rocks in Japan (Nagata et al, 1953; Kawai et al, 1954; Asami, 1954; Momose, 1958, 1963; Hirooka, 1961, 1963; Kawai, 1963). According to the results obtained by these investigators, the mechanism of the acquisition of the reverse N.R.M. should be classified into the two processes: (1) rocks had magnetized at the time of the reversal of the geomagnetic field, (2) rocks had acquired the reverse N.R.M. in the process of the self-reversal during or after the formation of these rocks. Among these studies, Momose (1958, 1963) discussed the mode of the variation of the geomagnetic dipole field from the normal to the reverse during the Pliocene. Hirooka (1963) and Kawai (1963) suggested the possibility of the periodic change of the geomagnetic dipole field from the normal direction to the reverse and back to the normal in the late Tertiary and the early Quaternary, the period being either 6 million years or 2 million years if it had existed. Doell and Cox (1961) reported that "the stratigraphic distribution of normally and reversely magnetized rocks studied palaeomagnetically strongly supports the field reversal hypothesis." In Iceland, Sigurgeirsson (1957) investigated the Tertiary lava flows with the intermediate directions of N.R.M. which were found only at a

boundary between normal and reverse groups. He suggested that the distribution of the magnetic poles corresponding to such intermediate directions represents the transition of the geomagnetic field from the normal to the reverse.

According to the above-mentioned investigations, it seems that the reversal of the geomagnetic field occurred several times in the late Tertiary and the early Quaternary and rocks with the intermediate directions appear to be one of examples to reveal the mechanism of the reversal, because it gives us an information of a change of the geomagnetic field when it moved from the normal to the reverse or in the opposite way.

In this paper, the author reports the results of measurements of the N.R.M. of Nose intrusive rocks, which show the normal and the reverse magnetization in one rock body, and these directions of magnetization show intermediate directions deviated from the present magnetic poles.

2) Geology and Sampling Sites

An intrusive rock mass of quartz diorite is exposed at Nose district in Osaka Prefecture, its maximum width being of 5.5 km and its length of 15 km (Sakaguchi, 1961). This rock mass was intruded into the upper Palaeozoic formation, and metamorphosed rocks already existed on the spot. This rock is partly covered by the Plio-Pleistocene sediments. Although the exact

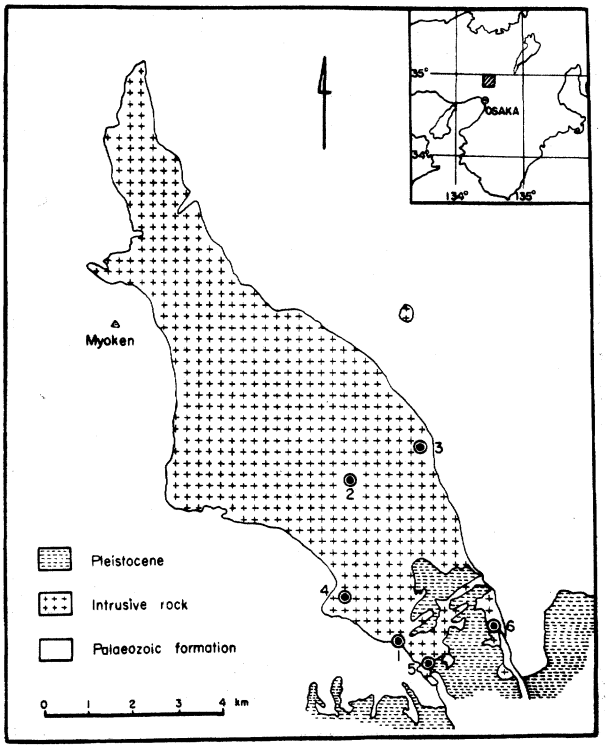


Fig. 1 Geological map of Nose district (after Sakaguchi). Circles show the sites where rock samples are collected.

age of intrusion is not known for the lack of geological evidence in this region, it is estimated to be the time between the late Mesozoic and the late Tertiary from the field survey of the same kind of intrusive rocks found in other localities in Kinki Province.

62 oriented samples of about 200 g in weight were collected from six localities as shown in Fig. 1 and also on Table 1. About 10 samples independently oriented were collected as widely as possible from each outcrop. Samples collected at the site 1 belong to the intrusive rock obtained at a contact zone, as seen in Fig. 1, while those collected at the site 2 seem to be situated at nearly the central part of the rock mass. Samples from other sites were also intrusive rocks collected at places slightly distant from the contact zone.

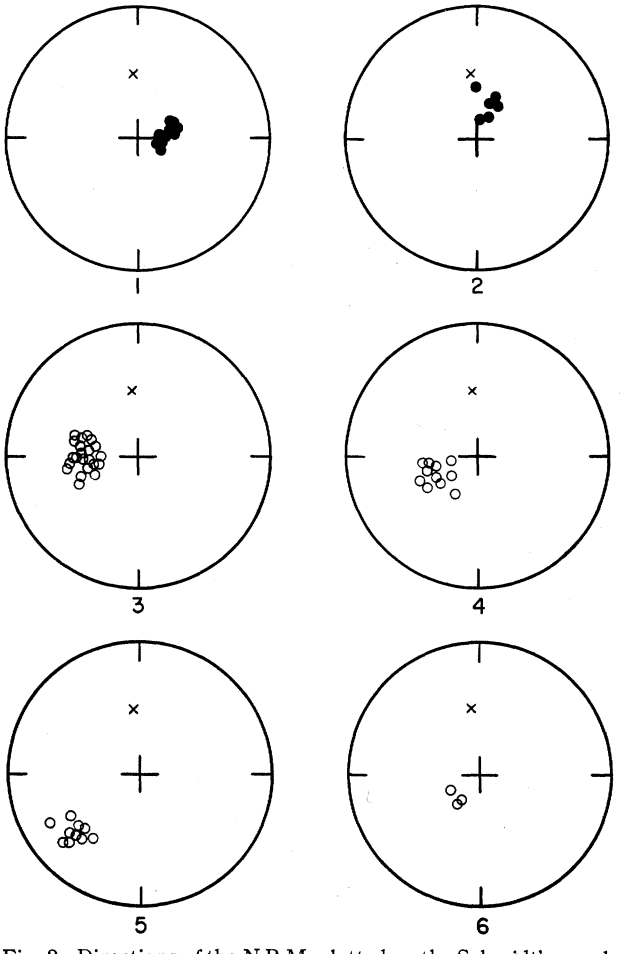


Fig. 2 Directions of the N.R.M. plotted on the Schmidt's equal area projection. Numbers 1 to 6 correspond to those of Fig. 1 and Table 1.
× : Direction of the present geomagnetic field
● : North-seeking directions being in the lower hemisphere
○ : North-seeking directions being in the upper hemisphere

Table 1

	Locality	Number of samples	Mean direction		Pole position		Intensity of N.R.M. (e.m.u./g)	Circle of confidence (5%)
			Decl.	Incl.	Lat.	Long.		
1	Baba	10	N82° E	+71°	32° N	177° E	4.10×10^{-5}	3.9°
2	Tarumi	6	N23° E	+66°	68° N	180° E	0.38	7.7°
3	Shimootoba	22	N90° W	−54°	19° S	16° E	5.37	3.5°
4	Umebara	11	S63° W	−60°	41° S	19° E	2.53	5.3°
5	Uemura	10	S50° W	−34°	43° S	52° E	0.67	5.7°
6	Kuwabara	3	S45° W	−68°	54° S	5° E	0.98	9.0°

3) Measurements and Results

The experimental process is the same given by Kawai (1951). The result of measurement of the N.R.M. is shown on Table 1. The direction of the N.R.M. of all samples is plotted in Fig. 2. In this figure, the inclination of the N.R.M. of the samples collected at the site 1 is in the lower hemisphere of the Schmidt's projection, and the declination deviates easterly compared with the present field direction. The direction of the N.R.M. of sedimentary rock baked by the intrusion is parallel to that of the intrusive rock. However, their intensity is weak (less than 10^{-6} e.m.u./g) and the detail of result is not discussed now. The declination of the N.R.M. of the samples collected at the site 2 is slightly eastward and the inclination is downward, but the magnetic direction at this site cannot be discussed in detail because of the following geological condition. This site may have been in contact with the country rocks which had covered the intrusive rock and the accurate distance between this site and the contact plane cannot be estimated. The inclinations of the samples collected at the sites 3, 4, 5 and 6 are upward, and the declinations are westward or southwestward, and they seem to be approximately antiparallel to the directions of the N.R.M. of the rocks in the sites 1 and 2.

The mean intensity of magnetization of all samples is of 2×10^{-5} e.m.u./g, and there is

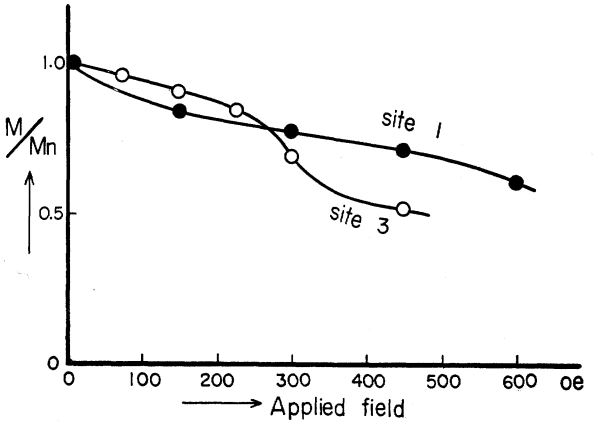


Fig. 3 Change in the intensity of the N.R.M. with progressive demagnetization by alternating magnetic field.
M/Mn: Normalized intensity of the N.R.M.
Mn : Initial intensity of the N.R.M.

no significant difference between the samples of the normal magnetization and those of the reverse one. The ratio J_n/J_i is very large as compared with that of ordinary granitic rock. One specimen from each site was demagnetized by the A.C. method. Fig. 3 shows two of the A.C. demagnetization curves as typical examples. The value of M/Mn decreases gradually with increasing applied field, being of about 0.5 for the A.C. field of 450–600 oe. The direction of magnetization remains completely unchanged, and this fact shows that these rocks have the very stable N.R.M..

4) Discussion

According to the data reported so far, the configuration of the geomagnetic field is approximated to that of a geocentric dipole in the past 35 million years (Doell and Cox, 1961). However, it is likely that the field had frequently changed the polarity during the Tertiary. On the other hand, the rocks magnetized in the intermediate directions of N.R.M. are found only at the strata between normally and reversely magnetized rocks (Sigurgeirsson, 1957). Several examples of rocks with the intermediate directions are reported in Iceland (Sigurgeirsson, 1957), Japan (Momose, 1958, 1963; Hirooka, 1961, 1963; Ito, 1963) and North America (Watkins, 1963).

The term of "the intermediate direction" is used for the direction of magnetization in the case when the magnetic pole is situated in latitudes lower than 55°.

Sigurgeirsson (1957) and Momose (1958) suggested that the transition of the geomagnetic field could be traced through successively from normal to reverse or reverse to normal, and they showed that the samples with the intermediate direction suggest the existence of transition period in the process of reversal. Hirooka (1963) pointed out a possibility of a periodic reversal of the field, because the distribution of the reverse N.R.M. in contemporaneous rocks all over the world shows a good consistency. But he could not trace the whole path corresponding to the transition of the geomagnetic field, because of unconformities above and below the rocks with the intermediate N.R.M.. Watkins (1963) found a succession of Miocene lava flows which show the intermediate normal and reverse directions of magnetization.

According to the above-mentioned investigations, the geomagnetic field in Tertiary is considered to have repeated the reversal either without changing the axis of the geomagnetic dipole field or with changing its axis along a certain route from time to time. If the rocks with the intermediate N.R.M. suffered no influnce from any tectonic movement or non-dipole field, they should have been formed at the time of the transition of the geomagnetic field, and these directions of the N.R.M. are expected to show a vestige of the change of the geomagnetic field.

The direction of the N.R.M. of the intrusive rock body in Nose deviates considerably from the axis of the present dipole field, and it shows an intermediate direction. At the contact zone, the direction of the N.R.M. of the intrusive rock is normal, while that of the intrusive rock at places not far from the contact zone is reverse. This fact is explained by the assumption that the geomagnetic field had been normal at the time when the rock at the contact zone was cooled down through the Curie point of ferromagnetic minerals contained

in them, but the field had reversed its polarity when the rock inside the intrusive mass was cooled down later.

Bearing in mind the above-mentioned facts, it appears that the direction of the N.R.M. of this intrusive rock indicates the position on a route, along which the pole moved in process of the reversal of the geomagnetic field. In Fig. 4, these positions are plotted on the Lambert's

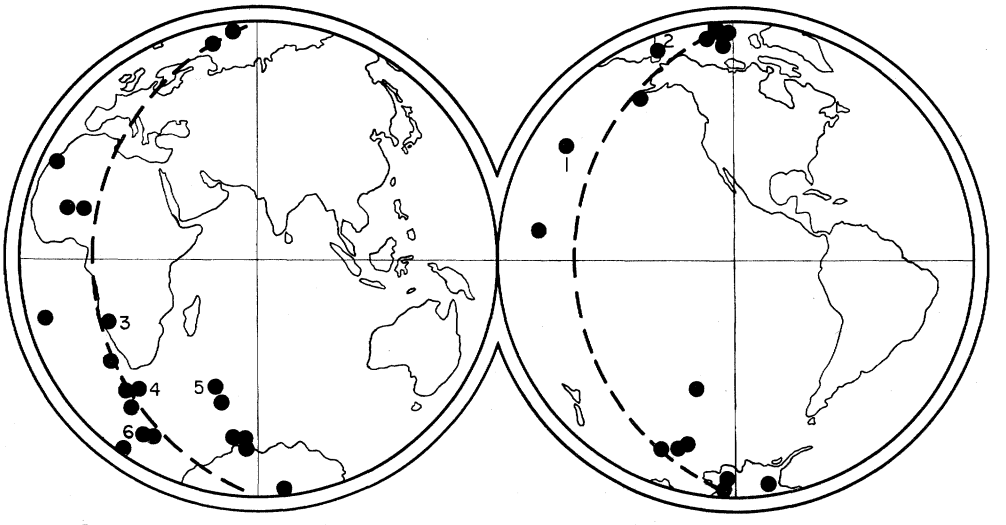


Fig. 4 Pole positions calculated from the Mio-Pliocene data. The numbers from 1 to 6 are corresponding to those in Figs. 1 and 2 and Table 1. Dotted line is a path of the transition of the magnetic pole inferred from them. The projection is Lambert's polar-equal net.

equal net. In the same figure, the pole positions which were estimated from the volcanic rocks of upper Miocene and lower Pliocene in Setouchi Province and the northern part of Tajima investigated by Hirooka (1961, 1963), and those of the Mio-Pliocene rocks in San-in and Kyushu Provinces obtained by the author (Ito, 1963), are also shown. The numbers 1, 2, 3, 4, 5 and 6 correspond to those in Figs. 1 and 2 and on Table 1. As seen in Fig. 4, the distribution of the pole positions which were examined on the Mio-Pliocene rocks in southwestern Japan is almost limited in the definite zone on the projection. A dotted line in the figure indicates a path of transition estimated from the distribution of the pole positions.

The occurrence of the intermediate N.R.M. can be explained by some other processes without considering any continuous movement of the dipole. For example, if it is assumed that the whole rock mass of intrusion in question is subjected to a clockwise horizontal rotation of land by about 60° after it has been magnetized along the direction of a geocentric dipole field at the time of formation, the direction of the N.R.M. become intermediate as seen in Fig. 5, even if the direction of the dipole field had been either parallel or antiparallel to the present one. However, it seems more plausible to consider that the dipole shifted from the north towards the south or oppositely along a path as seen in Fig. 4, giving the intermediate directions to some rocks, since we can find no geological evidences about any kind of block movement around the present intrusive mass. All of the pole positions obtained from rocks collected at different parts of southwestern Japan lie in a great circle surrounding

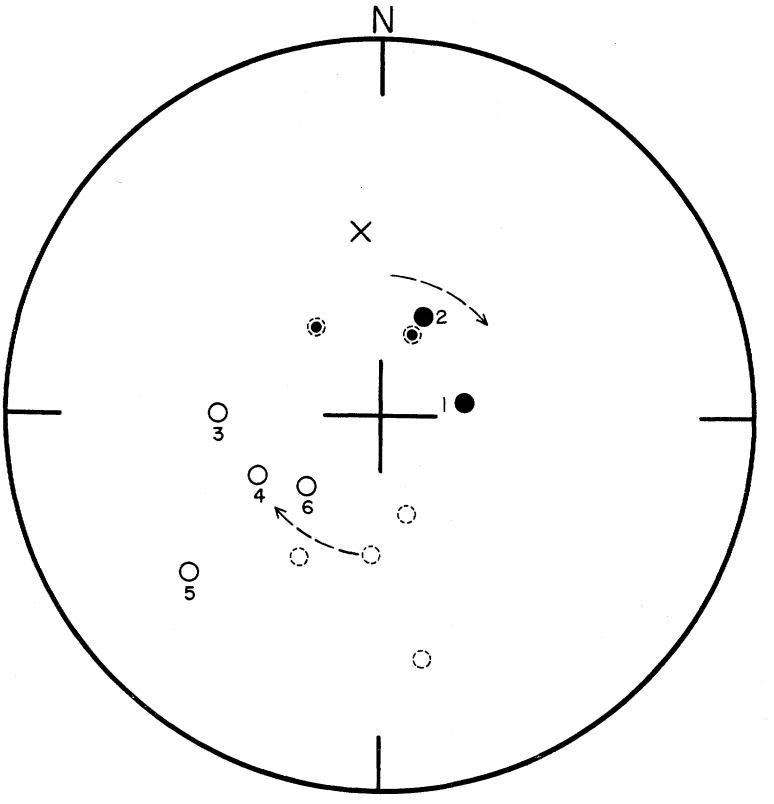


Fig. 5 Change of directions of the N.R.M. rotating clock-wisely in the horizontal plane of the rock body.
● ○ : Mean directions of the N.R.M. of each site
◉ ◌ : Assumed directions before the rotation
× : Direction of the present geomagnetic field

the earth, which includes the present pole with a very small deviation.

Taking the above-mentioned considerations into account, the following explanations can be given to the intermediate directions observed in the Nose intrusive mass.

The granitic rock collected at Nose acquired the N.R.M. at the time when the dipole was shifting from north to south, and consequently the directions of the N.R.M. of this intrusive mass are found to be intermediate. Therefore, using the directions of this N.R.M., the path of the transition of the dipole can be obtained. When we summarize the data of the N.R.M. of late Tertiary rocks collected from southwestern Japan, the whole path of the transition can be traced and some part of this path shows a fairy good agreement with that reported by Momose (1958), whose result has been obtained for the rocks in the central part of Japan. The time required for the transition from the normal magnetization to the reverse in the Mio-Pliocene is estimated to be of the order of 10^5 years on the basis of the difference of cooling rate between the contact zone and the places inside the intrusive mass of granite (Nishimura, 1961) and also the geological aspects of the sampling sites.

Acknowledgements

The author wishes to express his thanks to Professor N. Kawai of Osaka University for his valuable suggestions and advices in the course of the work.

Thanks are also due to Dr. S. Kume of Osaka University for many valuable suggestions and encouragements. He is indebted to Dr. K. Yaskawa of Fukui University and Mr. K. Hirooka of Osaka University for their help on the sampling and the measurements. His hearty thanks are due to Dr. T. Matsumoto of Osaka City University for his helpful advice on the geology in this region.

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